S2 Text. Audit and saved-result reproduction

This appendix separates computational provenance and operational preservation from scientific results. It is not an independent participant sample, a new experiment or proof that task labels isolate sustained motor imagery. The scientific definitions and new exploratory QC rules are in S1 Text.

Original scientific identities

The original CPU baselines, 54 source-only EEGNet fits and the single frozen checkpoint query evaluation remain distinct stages. The original accepted18 ledger was not rewritten when the separately accepted36 were appended. The combined source ledger has54unique jobs and SHA256 `013668c8d1b1c5d6061f3701d495cf7770743a1c6fda8678f5db7ffd50992670`. The seeds are20261012/20261013/20261014 for18published participant identities. Their number does not increase the independent participant count above18.

The original query operation used execution commit `d34703ca5bccf7a5b88a91c2ffb2cba893f7cec8`, contract `fb3a429776e77a4ee8de96c8c7f2a833a4d311cedd8f0a9bcec3d0c6e8df225a`. It used54original checkpoints, no optimizer or support update, and preserved model parameters, batch-normalization and source normalization states. Existing computational audits bound54archives,1,189inputs and the4290query identities. The original source54ledger is unchanged by the separate acceptance of these query results. This submission revision performs no model forward or fit.

Historical failure and preservation boundaries

The CSP diagnostic's first attempt stopped after the process monitor could not read a procfs interface; its claim, authorization, real exception and private backup were retained. An explicitly authorized run02 used the compatibility fix, retained the scientific consumer and source/support-only scope, and completed240prescribed SOURCE20pipeline attempts. This is not an automatic retry or a newly chosen scientific analysis. The former broad-except preparation script remains archived as superseded; its old synthetic test pass does not make it an authorized current entry point.

Two source-model batch controllers paused at off-machine receipt boundaries; their same-state resumes and original deadlines remain preserved in the full repository audit tree. Training, actual off-machine export, scientific audit, full private R2 GET verification and real parent acceptance were separate states. No training-complete flag was treated as acceptance. Historical false/pending producer flags and the actual failure logs were not silently rewritten.

The full repository retains these records. Journal files omit live host/session access details, credentials and the original weights/EEG packages. A signed-off receipt in this project is a recorded operational acknowledgement, not an independent human replication or a cryptographic signature unless explicitly identified as such. In particular, the actual query parent acknowledgement reports complete-receipt and result-identity review and saved arithmetic; it does not report another networkGET, model forward or complete original-byte rehash of all147objects. The operator's original fullGET verification remains separate from that subsequent review.

New saved-prediction QC lineage

The new post-observation protocol was recorded before its computation and retains the original250µV maximum-channel peak-to-peak flag. It maps25,740saved prediction rows to4290unique

included query events, retaining the30excluded partial query identities in the broader4320-row query ledger. All144original participant endpoint reconstructions matched saved results exactly. Public predictions and metadata were independently reconciled to their original CPU and EEGNet text sources without loading a model or EEG. Every source hash and mapping is recorded in the accompanying input manifests.

The public minimal inputs include all7200frozen event metadata rows, all saved method/seed predictions, the existing analysis configuration and original participant endpoints. The joined output includes sample/class counts and explicit missing-class states. No EEG samples, learned parameters or checkpoint archive are present. The non-numeric path strings in provenance records identify historical inputs; the public reproducer uses explicit portable paths instead.

Only10/12remaining participants have a complete unflagged five-session endpoint. The subset comparison therefore uses the same ten participants and does not mislabel a12-versus10difference as a paired QC effect. Pooled-query endpoints are separate because they weight sessions differently. The data dictionary records units, participant/session ordering, subset and method names and every missing value convention. No missing BA is recoded as zero or chance.

Reproducing paper arithmetic without EEG or weights

Unpack S1 Data and use the exact command in its REPRODUCE.md. The new `scripts/analyze_kumar_saved_query_qc_20261015.py` supports an explicit `--portable-inputs` path and a fresh `--output` directory. It reads saved text only and must not invoke any training or checkpoint-query launcher. An actual public-input replay reproduced all13scientific CSVs byte-for-byte. Separately implemented arithmetic and source-lineage checks also matched the joins, class eligibility, seed averaging, five-session weighting, percentile bootstrap and leave-one-participant-out results. These computational checks were themselves AI-assisted and do not replace the author's

final human review.

S1 Data also supplies all2871saved epoch history rows and54fit summaries. Their export uses training text, not deserialized weights. Thirty-seven console logs are empty, while all54JSON histories are complete; an empty log does not prove zero execution. Source/target metrics and every seed/collapse remain visible. The original archive audit hashes binary members without decoding them.

Exact figure data are supplied for all four figures. The new Fig4 contains 180participant/session contrast cells, preserving both cohorts and all negative values. Reproduction reads saved CSV/JSON and writes a new output directory; it neither reruns a scientific pipeline nor recomputes the original bootstrap intervals. Main-figure TIFFs are actual flattened600dpi RGB/LZW exports. Their physical dimensions and hashes are checked. The installed typeface is Liberation Sans rather than a claim of installed Arial; this remaining production-font choice is noted in the author worksheet.

Availability, licenses and scope

Public source EEG is obtained from the exact NEMAR version cited in the main paper, under its original terms. Private R2 preserves original experiment packages, but is not a public data endpoint. The paper-specific public allowlist contains saved predictions/metrics, QC metadata, graph data, original code and reproduction instructions, not unrelated datasets or complete software wheels. The proposed MIT-original-code and CC BY4.0-derived-content choices remain for the author's selection, with third-party notices preserved separately. CSL CC BY-SA source and noncommercial optional software are not included as journal article content. No archive DOI or journal release is invented.

The final artifact manifest binds the source commit, exact file lengths and SHA256s. The current revision does not alter prior scientific configurations, accepted ledgers, partial-window exclusions

or failure evidence. Any unavailable new R2 synchronization is reported as a transport limitation and is not replaced by a fabricated fullGET receipt. The fixed Git version and local document paths remain independently inspectable. This task does not submit the manuscript.